

A Toy Model of Global Mitigation Incentives

Working Draft

1. National Level: The Investment Choice

1.1 The one equation that matters

In a putty-clay world, almost all abatement comes from the choice of new capital. The central variable is ϕ_t : the **clean share of new investment** in year t .

$$\phi_t = \frac{1}{1 + \exp\left(\beta \left[\underbrace{c_t^C}_{\text{clean cost}} - \underbrace{c_t^D}_{\text{dirty cost}} \right]\right)} \quad (1)$$

This is a logit. When clean is cheaper ($c^C < c^D$), $\phi \rightarrow 1$. When dirty is cheaper, $\phi \rightarrow 0$. The parameter β captures how price-sensitive the investment decision is: high β means a sharp switch near cost parity (power sector utilities); low β means a gradual transition with a wide distribution of tastes (household vehicle choice).

This is the CES in disguise. The logit share model and the CES give nearly identical results; the logit is just more transparent for a binary clean/dirty choice.

1.2 What goes into the costs

$$c_t^C = \underbrace{k_t^C \cdot a(r_t)}_{\text{annualised capital}} + \underbrace{p_t^C / \eta^C}_{\text{fuel per unit service}} - \underbrace{\tau_t \cdot (e^D - e^C)}_{\text{carbon price advantage}} - \underbrace{s_t}_{\text{subsidy}} \quad (2)$$

$$c_t^D = k_t^D \cdot a(r_t) + p_t^D / \eta^D \quad (3)$$

where k is capital cost, $a(r)$ is the annualisation factor at discount rate r , p/η is fuel cost per unit of service (price divided by efficiency), τ is the effective carbon price (from any source: tax, ETS, standard, CBAM), and s is any clean subsidy (CfD premium, purchase rebate, etc.).

The cost gap that drives the logit is:

$$\Delta_t \equiv c_t^C - c_t^D = \underbrace{(k_t^C - k_t^D) \cdot a(r_t)}_{\text{capital gap}} + \underbrace{(p_t^C / \eta^C - p_t^D / \eta^D)}_{\text{fuel gap}} - \underbrace{\tau_t (e^D - e^C)}_{\text{carbon price}} - \underbrace{s_t}_{\text{subsidy}} \quad (4)$$

Every national policy instrument enters through Δ :

- Carbon tax / ETS $\rightarrow$ increases τ
- CfD / RE auction $\rightarrow$ reduces c^C (guaranteed revenue = lower effective k^C)
- Vehicle standard $\rightarrow$ acts as a mandate on ϕ directly (overrides the logit for regulated shares)
- Subsidy $\rightarrow$ reduces s (directly narrows Δ)
- Fuel excise $\rightarrow$ increases p^D (widens fuel gap)
- MDB derisking $\rightarrow$ reduces r , which reduces $a(r)$, which narrows the capital gap

1.3 Capital stock evolution

The stock evolves as:

$$S_t^C = (1 - \delta) S_{t-1}^C + \phi_t \cdot I_t \quad (5)$$

where δ is the retirement rate and I_t is total new investment (replacement + growth). The clean share of the operating stock is $w_t^C = S_t^C / (S_t^C + S_t^D)$. Emissions are:

$$E_t = (1 - w_t^C) \cdot Q_t \cdot \frac{e^D}{\eta^D} \quad (6)$$

where Q_t is total service demand. This is the entire national emissions model: the logit determines new investment shares, the vintage accounting determines the stock, and the stock determines emissions.

1.4 The TECP in this model

The TECP is the value of τ that, acting alone (with $s = 0$), would produce the same ϕ_t as the actual policy mix:

$$\tau^* = \frac{\Delta_{\text{baseline}} - \Delta_{\text{policy}}}{e^D - e^C} \quad (7)$$

where Δ_{baseline} is the cost gap without any policy and Δ_{policy} is the cost gap with all instruments. This collapses the entire multi-instrument portfolio into one number.

2. Global Level: Leader–Follower Diffusion

2.1 The two equations that matter

The global model has two state variables: the **global technology cost** k_t^C (driven by cumulative deployment) and the **clean investment share in each country** $\phi_{i,t}$ (driven by local costs, which depend on the global technology cost plus local factors).

Global learning curve (technology cost falls with cumulative deployment):

$$k_t^C = k_0^C \cdot \left(\sum_i \sum_{\tau \leq t} \phi_{i,\tau} I_{i,\tau} \right)^{-\gamma} \quad (8)$$

where γ is the learning elasticity ($\gamma \approx 0.3$ for solar, meaning cost falls $\sim 20\%$ per doubling of cumulative capacity). The sum runs over all countries — deployment anywhere reduces costs everywhere.

Country-level adoption (each country's clean share responds to local cost gap):

$$\phi_{i,t} = \frac{1}{1 + \exp\left(\beta_i \left[k_t^C \cdot a(r_i) + p_i^C / \eta^C - p_i^D / \eta^D - \tau_i (e^D - e^C) - s_i - g_i \right]\right)} \quad (9)$$

where g_i is the international augmentation: the effective cost reduction from MDB guarantees, concessional finance, CfDs backed by international support, and other instruments that reduce r_i or provide s_i externally.

These two equations create the leader–follower dynamic:

1. **Leaders** (large τ_i , large s_i , low r_i) adopt early $\rightarrow$ high $\phi_i \rightarrow$ large contribution to cumulative deployment $\rightarrow k^C$ falls globally.
2. **Followers** (small τ_i , small s_i , high r_i) adopt later, but adoption is easier because leaders have driven k^C down.
3. **International augmentation** (g_i) accelerates follower adoption by closing the residual cost gap after the leader-driven technology cost reduction.

2.2 Where international instruments enter

Each international instrument enters through a specific channel in the country-level equation:

Instrument	Channel	Effect on Δ_i
Leader deployment (China solar, US IRA)	k_t^C (learning curve)	Reduces capital cost for everyone
MDB guarantee / concessional lending	r_i (discount rate)	Reduces annualised capital cost in country i
CfD backed by international finance	s_i or r_i	Guaranteed revenue reduces effective cost
CBAM	τ_i (effective carbon price)	Creates carbon price in exporting country
Subsidy reform via MDB DPO	p_i^D (dirty fuel price)	Removes subsidy, raises dirty cost
Carbon market credit purchase	s_i (revenue transfer)	Direct payment for clean deployment
Technology transfer / co-financing	k_t^C or s_i	Reduces clean capital cost locally

2.3 The tipping point

The tipping point is when $\Delta_i = 0$: clean cost equals dirty cost in country i . After this point, $\phi_i > 0.5$ and the transition is self-sustaining. The global model predicts an S-curve of adoption in each country, with the inflection point (tipping) occurring at different times depending on local costs and policies:

$$t_i^{\text{tip}} : k_t^C \cdot a(r_i) + p_i^C / \eta^C = k^D \cdot a(r_i) + p_i^D / \eta^D + \tau_i(e^D - e^C) + s_i + g_i \quad (10)$$

Countries with low r_i , high τ_i , and high g_i tip first. Countries with high r_i , low τ_i , and no g_i tip last. The time gap between first and last tipping is the “diffusion lag” — the central variable for global climate outcomes.

Policy objective: **minimise the diffusion lag**. International instruments (g_i , CBAM on τ_i , MDB on r_i) compress the lag by pulling followers forward.

2.4 The global TECP

For the global system, define the **global TECP** as the uniform carbon tax that, applied to all countries simultaneously (with no other instruments), would produce the same global emissions path:

$$\tau_{\text{global}}^* : \sum_i E_i(\tau_{\text{global}}^*) = \sum_i E_i(\text{actual policy mix}) \quad (11)$$

This is a numerical inversion (not a formula), because the learning curve makes the response nonlinear: a global carbon tax that drives early deployment also drives cost reductions that amplify later deployment. The learning externality means the global TECP is *less than* the sum of country TECPs — the interaction between deployment and cost reduction means the policy instruments are more powerful collectively than individually.

3. The Complete Model on One Page

Three equations, six parameters, everything else follows.

$$\text{Investment choice: } \phi_{i,t} = \frac{1}{1 + \exp(\beta_i \cdot \Delta_{i,t})} \quad (\text{adoption})$$

$$\text{Cost gap: } \Delta_{i,t} = (k_t^C - k^D) \cdot a(r_i) + (p_i^C / \eta^C - p_i^D / \eta^D) - \tau_i(e^D - e^C) - s_i - g_i \quad (\text{incentives})$$

$$\text{Learning curve: } k_t^C = k_0^C \cdot \left(\sum_{i,\tau \leq t} \phi_{i,\tau} I_{i,\tau} \right)^{-\gamma} \quad (\text{technology})$$

Six parameters: β_i (price sensitivity of investment, by country/sector), r_i (discount rate, by country), γ (learning rate, by technology), δ (capital retirement rate, by sector), $\eta^{C/D}$ (efficiency, by technology), $e^{C/D}$ (emission factor, by fuel). All observable or calibratable from data.

Policy instruments: τ_i (carbon price from any source), s_i (domestic subsidies), g_i (international augmentation), r_i (reducible by derisking). Plus mandates that override ϕ directly.

State variables: k_t^C (global clean technology cost, shared), $S_{i,t}^C$ and $S_{i,t}^D$ (national capital stocks, by vintage).

Output: $E_{i,t}$ (national emissions) and τ_i^* (national TECP).

Everything in the previous 20 pages of tables reduces to these three equations. The national model is equations (adoption) + (incentives). The global model adds (technology). The TECP is the τ that makes (incentives) produce the observed ϕ without s or g . The tipping point is $\Delta = 0$. The diffusion lag is the time between the first and last country crossing $\Delta = 0$.